

Lab 10, loop analysis

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Step 1) Write the output of the following code (Do not use any software, including Visual Studio code, to run the code yet. You will test the code later in step 3):

a)

```
int sum=0, count=0;
for (int n = 20; n> 0; n-=2){
    if (n%3 ==0){
        count++;
        continue;
    }
    sum += n;
}
cout<<"The sum is: "<<sum<<endl<<"The count is: "<<count<<endl;
```

Loop iteration	n	if (n%3 ==0) True or false	count	sum
1	20	false	0	0+20=20
2	18	true	0+1=1	20
3	16	FALSE	1	20+16=36
4	14	FALSE	1	36+14=50
5	12	TRUE	1+1=2	50
6	10	FALSE	2	50+10=60
7	8	FALSE	2	60+8=68
8	6	TRUE	2+1=3	68+4=72
9	4	FALSE	3	72
10	2	FALSE	3	72+2=74
11	0 STOP FOR LOOP			

b)

```
int i=0, add=0;
while(i<=10){
    i++;
}
```

```

    if(i<5 && i !=2){
        cout<<"i = "<<i<<endl;
    }
    else{continue;}
    add +=i;
}
cout<<"Add = "<<add<<endl;

```

Loop iteration	i	if(i<5 && i !=2) True or false	else True or false	add
1	0+1=1	True	False or skipped	0+1=1
2	1+1=2	FALSE	True → continue	1
3	2+1=3	True	False or skipped	1+3=4
4	3+1=4	True	False or skipped	4+4=8
5	4+1=5	FALSE	True → continue	8
6	5+1=6	FALSE	True → continue	8
7	6+1=7	FALSE	True → continue	8
8	7+1=8	FALSE	True → continue	8
9	8+1=9	FALSE	True → continue	8
10	9+1=10	FALSE	True → continue	8

Step 2) Discuss your result from step 1 with your partner.

Step 3) After the discussion write the code in Visual Studio Code and run the two codes. Check the output of the code and compare it with your output from step 1. Write which is the mistake, if any, that you made from step 1.